

Assignment: Semiconductor Devices

1. What is the difference between conductor, semi-conductor and insulator?
2. Which semiconductor material used mostly for IC design and why?
3. What is the difference between Silicon and Germanium?
4. What is the difference between intrinsic and extrinsic semiconductor?
5. What is doping? What are the different types of trivalent and pentavalent impurity materials?
6. What is the difference between drift and diffusion?
7. What is PN junction? How it behaves without any external bias voltage?
8. What is built-in potential of a P-N junction?
9. What is the PN Junction behavior at forward bias as well as reverse bias?
10. What is reverse saturation current of a PN junction diode?
11. What is junction or diffusion capacitance of a reverse biased PN junction?
12. Draw the V-I characteristics of PN junction diode.
13. What is BJT and how it is different from the diode?
14. What is the difference between NPN and PNP BJT?
15. What are the working principles of BJT?
16. Draw the input and output characteristics of BJT
17. What is MOSFET and how it is different from BJT?
18. What are the different types of MOSFET?
19. What are the different terminals of MOSFET?
20. What is the difference between P-channel and N-channel MOSFET?
21. Why MOSFET is suitable for IC design?
22. Define the working principle of MOSFET?
23. What are three different region of operation of a MOSFET?
24. Draw the I_d - V_{gs} and I_d - V_{ds} characteristic curve for MOSFET.
25. What are the different intrinsic capacitances associated with MOSFET?